

**KATHMANDU UNIVERSITY**  
**SCHOOL OF ENGINEERING**  
**DEPARTMENT OF MECHANICAL ENGINEERING**



**PROJECT REPORT ON**  
**Inspection and Preliminary Maintenance of CNC VESTA 660 at KU**

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2025-11-09

Date: 2025-11-09

To,  
The HOD  
Department of Mechanical Engineering  
Kathmandu University

**SUBJECT: Cover Letter for report approval**

Dear Sir,

We hereby submit for your kind review of our report entitled “**Inspection and Preliminary Maintenance of CNC VESTA 660 at KU**”, prepared as part of the ongoing maintenance and repair activities of the CNC machine located at the Technical Training Center (TTC).

This report includes a comprehensive overview covering the **introduction, objectives, methodology, compartmental breakdown, examination conducted, results, and the conclusion** derived from the proposed inspection and preliminary maintenance work.

We respectfully request your thorough review and look forward to your continued support, guidance, and formal approval to proceed further with the necessary activities

Sincerely,

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## **LIST OF ABBREVIATIONS**

KU Kathmandu University

SOE School of Engineering

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# 1. INTRODUCTION

## 1.1. Background

A CNC machine is a motorized maneuverable tool and often a motorized maneuverable platform, which are both controlled by a computer, according to specific input instructions. Instructions are delivered to a CNC machine in the form of a sequential program of machine control instructions such as G-code and M-code, and then executed. The program can be written by a person or, far more often, generated by graphical computer-aided design (CAD) or computer-aided manufacturing (CAM) software. CNC offers greatly increased productivity over non-computerized machining for repetitive production, where the machine must be manually controlled or mechanically controlled by pre-fabricated pattern guides.



Figure 1: CNC at KU

The **Hwacheon VESTA-660**, a **3-axis vertical machining center**, was procured in 2014 to enhance practical learning in the mechanical engineering laboratory at Kathmandu University. The machine operates along three linear axes (X, Y, and Z), enabling it to perform high-precision milling, drilling, and contouring operations on a wide range of materials. It features a robust and compact structure designed to ensure high-speed, stable cutting. The VESTA-660 is equipped with a spindle speed of up to 10,000 RPM, supporting high-speed machining of intricate components.

Its working table measures 850 mm × 410 mm, with a maximum table load capacity of 300 kg. The X-axis travel is 660 mm, Y-axis travel is 410 mm, and Z-axis travel is 460 mm, making it suitable for medium-sized parts and precision machining tasks commonly required in both industrial and educational contexts.

From the time of installation, it served as a vital tool for undergraduate students, enabling hands-on experience in precision machining, programming, and automation systems. The machine was actively used from **2014 to 2017** under the supervision of Rajkumar Thakuri Sir, who was primarily responsible for operating and instructing students on its usage. **After 2017**, the machine **was not in active use** due to various circumstances. Later, during the nationwide lockdown in 2019, all academic and technical activities were suspended. During this time, due to unavoidable circumstances, routine laboratory cleanliness and maintenance could not be carried out by the administration.

As a result, the machine remained inactive for a prolonged period, creating an environment that allowed rodent intrusion. Over time, mice severely damaged internal wiring, especially within the operating panel and control circuits, leading to the complete breakdown of the system.

Although there were proposals to bring in foreign service providers for the repair and maintenance of the CNC machine, a team of students expressed the desire to undertake the repair process themselves. The objective was to both know the root cause of breakdown of the machine and gain practical experience in CNC diagnostics and repair.

This report presents the findings, repair progress, and insights gained during the student-led restoration of the CNC VESTA 660

## **1.2. Objectives**

- To gain practical knowledge and hands-on experience in CNC inspection and Preliminary Maintenance during the process
- To diagnose the main reason behind the failure of the CNC VESTA 660 machine.
- To inspect the CNC machine and identify both damaged and working parts.

## **1.3. Significance**

- The CNC machine is an important tool for mechanical engineering students to learn practical machining and automation.

- Repairing the machine helps continue hands-on learning activities that were stopped due to its breakdown.
- The repair process itself provides a valuable learning opportunity for students involved in troubleshooting and fixing real-life problems.
- Bringing the CNC machine back to working condition saves the cost of buying a new one.

#### 1.4. Limitations

- Some internal parts of the machine were hard to inspect without fully disassembling it.
- Without fully completing the wiring repairs, it is not yet possible to diagnose the functionality of the mechanical section, such as motor movement and spindle operation.

## 2. METHODOLOGY

### 2.1. External Components of the CNC VESTA 660

The CNC VESTA 660 is equipped with several external components that work together to ensure safe, precise, and efficient machining operations. Each component has a specific role but is interdependent with others for the complete functioning of the machine.

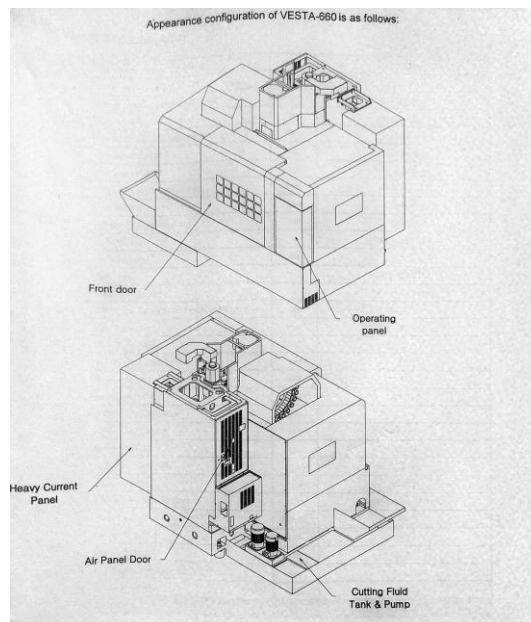


Figure 2: Machine Specification

### 1. Operating Panel



Figure 3: Operating Panel

The operating panel acts as the central control interface for the machine operator. It allows the user to input machining programs, adjust settings, initiate operations, and monitor machine status. Commands given through the panel directly control motor movements, spindle speed, coolant flow, and other systems. It is electronically linked with all major control and power units of the CNC.

## 2. Front Door

The front door serves both as a physical access point to the machining zone and as a safety enclosure. It is typically integrated with an interlock system that communicates with the control panel and power units to ensure that the machine does not operate while the door is open. This ensures user safety during tool changes or setup.

## 3. Heavy Current Panel



Figure 4: Heavy Current Panel/PSU

This panel houses the machine's high-voltage electrical components such as power relays, servo drives, contactors, and transformers. It is responsible for distributing electrical power to all active components, including motors, spindle, coolant pump, and lubrication system. It connects directly to both the operating panel (for command input) and the machine body (for mechanical output).

#### **4. Air Panel Door**

The air panel contains pneumatic components like solenoid valves, filters, and pressure regulators. These components manage air-driven functions such as tool clamping, spindle blowing, or chip removal. The pneumatic system is activated through the CNC's electrical control, making it closely connected to the heavy current panel and operating panel.

#### **5. Lubrication Tank**

The lubrication tank stores and supplies oil to moving mechanical parts such as linear guideways and ball screws. It ensures smooth operation and reduces wear and tear. The tank has sensors and a motorized pump, which are monitored and controlled via the control panel. The lubrication process is automated and synchronized with machine operation.

#### **6. Cutting Fluid Tank and Pump**

This system supplies coolant during machining to reduce heat, prevent tool wear, and wash away chips. The coolant pump is powered through the heavy current panel and activated by signals from the control system. It is functionally connected to the machining cycles and typically works alongside the spindle and feed systems.

These components together form an integrated electromechanical system. The operating panel acts as the command center, while the heavy current panel distributes power to various subsystems. Pneumatic and lubrication systems ensure mechanical reliability, and safety components like the front door prevent accidental operation. All components must function harmoniously to maintain the CNC machine's precision, safety, and performance.

### **2.2. Selected Part for Detailed Examination**

Among all external compartments, the operating panel of the CNC VESTA 660 was selected for detailed examination. This decision was based on the observation that the majority of damaged wires were concentrated in this section. The operating panel serves as the central

interface for communication between the machine and the operator and is critical for initializing, controlling, and monitoring machine operations.

The operating panel comprises the following three key components:

### **1. LCD/CNC Control unit**

The LCD / CNC Control Unit serves as the primary visual interface and houses the central processing "brain" of the machine. Its purpose is to execute the CNC software, manage complex motion control algorithms, and provide the operator with real-time feedback on the machine's status. It processes all system data, including axis position coordinates, spindle speeds, active G-codes, and diagnostic alarms.

Beyond display functions, this unit acts as the master communication hub, coordinating signals between the MDI input, the I/O modules, and the servo drives. By successfully passing the internal RAM/ROM integrity checks during diagnostic testing, this unit has demonstrated that its core logic and system memory are functional, confirming its ability to manage the machine's operations once external communication links are restored.

### **2. HMI(Human Machine Interface)**

The Human-Machine Interface (HMI) is the complete control interface between the operator and the CNC machine. It consists of various components including the rotary knobs, selector switches, status indicators, emergency stop, and control buttons for spindle, coolant, feed override, and axis jog. The HMI enables the operator to monitor machine operations, navigate menus, adjust process parameters, and control auxiliary systems like lubrication and cooling. It acts as the central hub for real-time interaction and supervision of machine functions.

### **3. MDI (Manual Data Input) Unit/Keyboard**

The Manual Data Input (MDI) unit is a specific mode within the HMI that allows the operator to manually enter individual G-code or M-code commands directly into the CNC controller. This is typically used for quick testing, setup movements, or simple operations without running a complete part program. The MDI mode utilizes the keyboard or touch input on the HMI panel, and is essential for efficient manual control during setup, troubleshooting, or calibration.

These three components are interconnected through internal wiring and circuit boards. Any damage in their wiring—especially from external sources such as rodents—can lead to system-

wide failures in user interaction, machine control, and command execution. As such, the operating panel was the logical focus for detailed inspection and repair during this project.

### **2.3. Activity Performed**

As part of the detailed inspection and repair initiative, preliminary maintenance work was undertaken on the CNC Hwacheon VESTA-660 equipped with the FANUC Series Oi-MD control system. The primary objective of this activity was to assess the overall condition of the machine and identify whether the observed malfunctions were due to external wiring issues or internal hardware failure within the control system and its peripheral components.

Prior to conducting electrical or functional testing, significant rodent debris, metal particles (chips), and accumulated dust were observed inside the machine's control panel and enclosure. Upon request, personnel from the Technical Training Center (TTC) carried out a thorough cleaning of the internal compartments to create a safe and workable inspection environment. The inspection was conducted under the technical supervision of Mr. Sameer Timsina, Production engineer currently working at Yatri Motorcycle, who provided expert guidance during the disassembly and evaluation process.

Following the cleaning procedure, the following activities were performed:

#### **2.3.1. External Casing Removal and Physical Inspection**

The external casing of the control system was carefully removed to allow access to all key components. A visual and physical inspection was conducted on all accessible parts, including:

- Operating Panel
- Power Supply Unit (PSU)
- Servo and Spindle Drive Modules
- Fan Units and Cable Harnesses
- Motors (visually, without electrical tests)

This inspection was done to identify visible signs of damage, corrosion, burnt wiring, or loose components. No mechanical damage or failure was detected in major subsystems aside from the operating panel.

### 2.3.2. Diagnostic Verification of the Fanuc 0i-MD Control & Display Unit

To assess the integrity of the **Fanuc 0i-MD Control Unit** and the LCD Display, a controlled isolation test was performed. Due to extensive rodent damage identified in the primary power loom, the internal **24VDC feed (Connector CP1)** was bypassed to prevent potential short circuits from reaching the main processing boards.

An external regulated 24V DC power supply was applied directly to the control unit's power input. The system successfully initiated the **Fanuc Boot System**, completing both **RAM and ROM integrity tests** with an "END" status. This result confirms that:

- **Hardware Viability:** The main CPU board, DRAM, and SRAM modules are electronically functional and do not require replacement.
- **Display Integrity:** The LCD backlight and screen module are operational, indicating that the failure to boot previously was a result of external wiring damage rather than hardware failure.
- **Isolation of Faults:** The diagnostic confirms that the current system "hang" at the boot loader is a logical issue (loss of SRAM data or missing I/O link signals) rather than a physical hardware defect in the display unit.



Figure 5: Display

### 2.3.3. HMI Board and MDI Condition Check

A detailed inspection of the Human-Machine Interface (HMI) was conducted, focusing on the Manual Data Input (MDI) unit and its internal interface circuitry. The following conditions were documented:

- **MDI Logic Interface:** The MDI Interface Board (A20B-2004-0430/01A) showed visible damage to the connector headers. Due to the proprietary nature of Fanuc's serial communication protocols, this unit must be restored to its original specification to ensure the CNC receives accurate input signals.
- **Internal Interconnect Failure:** The 40-pin MDI-to-CPU ribbon cable has suffered total failure due to severed conductors. Since this specific Fanuc model does not support standard USB keyboard inputs for system-level diagnostics, the restoration of this proprietary link is a critical path for machine recovery.
- **Signal Routing (JA3/CP1):** Severe degradation was found in the external wiring routed to the JA3 (Signal) and CPD1 (Power) ports. These links are essential for the MDI unit to communicate with the main CNC rack in the electrical cabinet.



Figure 6: Damaged parts

### 3. Replacement Required

The following components and cable assemblies have been identified as non-functional based solely on the current **diagnostic inspection and physical results**. Because these units handle proprietary Fanuc Serial Bus (FSSB) and I/O Link protocols, they must be replaced with specified industrial parts to ensure system integrity and allow the restoration process to move beyond the "Boot System" stage.

**Note:** This list represents only the damage identified during the preliminary phase; additional replacements may be required following the successful restoration of these core communication links.

| <b>Title</b>                                      | <b>Connector1</b>        | <b>Connector2</b> | <b>Cable</b>                         |
|---------------------------------------------------|--------------------------|-------------------|--------------------------------------|
| CPD1                                              | tab housing: 1-178288-3  | Ring Terminal     | OLFLEX-110 (tab housing: 1-178288-3) |
| CP1A                                              | tab housing: 1-178288-3) | Ring Terminal     | OLFLEX-110 (tab housing: 1-178288-3) |
| JA3                                               | Spade Terminal           | PCR-E20F          | OLFLEX-110                           |
| JD1A                                              | PCR-E20F                 | PCR-E20F          | LiTy-TP                              |
| JD1B                                              | PCR-E20F                 | PCR-E20F          | LiTy-TP                              |
| Programmable Controller CP1L I/O Connection Cable |                          |                   |                                      |
| CD38A                                             |                          |                   | Ethernet Cable                       |
| CN1                                               |                          |                   | Ribbon Cables                        |
| CN2                                               |                          |                   | Cables                               |
| CP1A                                              |                          |                   |                                      |
| CK27,CE54,CE53                                    |                          |                   |                                      |
| HIF3BB-50PA-2.54DSA (*2)                          |                          |                   |                                      |
| HIF3BA-30PA-2.54DSA                               |                          |                   |                                      |
| JA2,JA3,JA40                                      |                          |                   |                                      |
| JD51A                                             |                          |                   |                                      |

|                          |  |  |  |
|--------------------------|--|--|--|
| JD36A,JD36B              |  |  |  |
| F1, GND2                 |  |  |  |
| KEM CODE SWITCH GFED CBA |  |  |  |
| JA40                     |  |  |  |

## 4. RESULTS AND OBSERVATIONS

Based on the above activities, the following results and conclusions were drawn:

- CNC Control & Display Viability:** The Fanuc 0i-MD Control Unit and LCD monitor are electronically functional. Isolated testing via a 24VDC external supply confirmed that the main CPU board successfully passes internal **RAM and ROM integrity tests**. This suggests that the core "intelligence" of the machine is intact and the primary issues are restricted to external communication and power delivery.
- HMI and MDI System Integrity:** The Human-Machine Interface (HMI), specifically the MDI (Manual Data Input) unit, was identified as a critical failure point. Physical inspection revealed severed proprietary ribbon cables and terminal damage to the A20B-2004-0430/01A Interface Board.
- Absence of Peripheral Workarounds:** Research confirmed that this Fanuc 0i-MD model does not support standard USB keyboard inputs for system-level diagnostics or navigation. Therefore, the restoration of the proprietary MDI hardware is the mandatory "bottleneck" for all subsequent software diagnostics and parameter recovery.
- Cabling and Communication Faults:** Extensive rodent damage was documented on the JD1A/JD1B (I/O Link) and JA3 (Signal) cables. These faults currently prevent the CNC from communicating with its "limbs" (I/O modules and motors). Replacement is required to meet Fanuc's electromagnetic compatibility (EMC) standards.

- **Mechanical and Power Components:** Preliminary visual inspection of the Power Supply Unit (PSU), Servo Motors, and Spindle Drive did not reveal immediate signs of catastrophic failure. However, a definitive status for these high-voltage components can only be established once the MDI is restored and the system can be taken out of "Boot Mode".
- **Environmental Remediation:** The interior of the control cabinet, previously compromised by rodent activity and nesting debris, has been professionally cleaned and decontaminated. This has stabilized the hardware environment to prevent further corrosion or short-circuiting during the restoration phase.

## 5. CONCLUSION

The preliminary inspection and diagnostic phase of the **Hwacheon Vesta 660** restoration have yielded a high degree of confidence in the machine's core electronics. The successful power-on test of the **Fanuc 0i-MD Control Unit** confirms that the central processor and memory modules are functional, effectively preserving the most expensive hardware components of the system.

However, the restoration of full operational capability is currently obstructed by physical damage to the Human-Machine Interface (HMI). Specifically, the destruction of the MDI unit's proprietary ribbon cables and the A20B-2004-0430/01A Interface Board prevents any navigation beyond the initial boot screen. As research has confirmed that this model does not support auxiliary input devices like USB keyboards, the procurement and installation of these specific Fanuc components are mandatory for project advancement.

While the current results focus on the communication and control bottlenecks identified to date, the team anticipates a phased approach to the remaining electromechanical subsystems. Once the MDI and I/O Link (JD1A/JD1B) cables are restored, we will proceed with a comprehensive verification of the Servo system and Spindle drive against the original factory accuracy standards.

In summary, the project has moved from a state of total inactivity to a verified "functional brain" status. By addressing the specific proprietary hardware needs identified in this report, the Department can proceed toward a cost-effective and technically sound recovery of this critical laboratory asset.



## APPENDIX

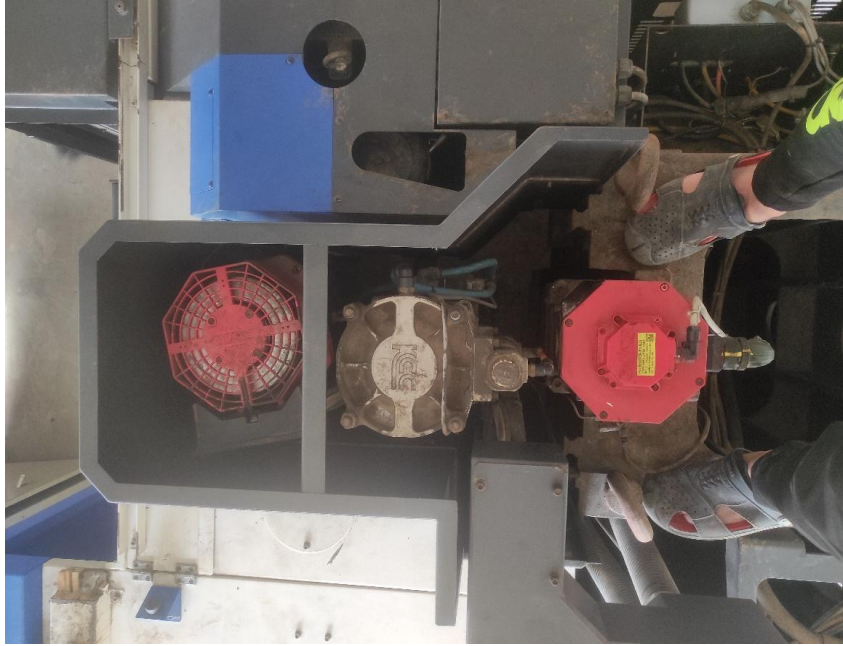


Figure 7: Top View of machine



Figure 8: Processing Unit



Figure 9 : HMI part I

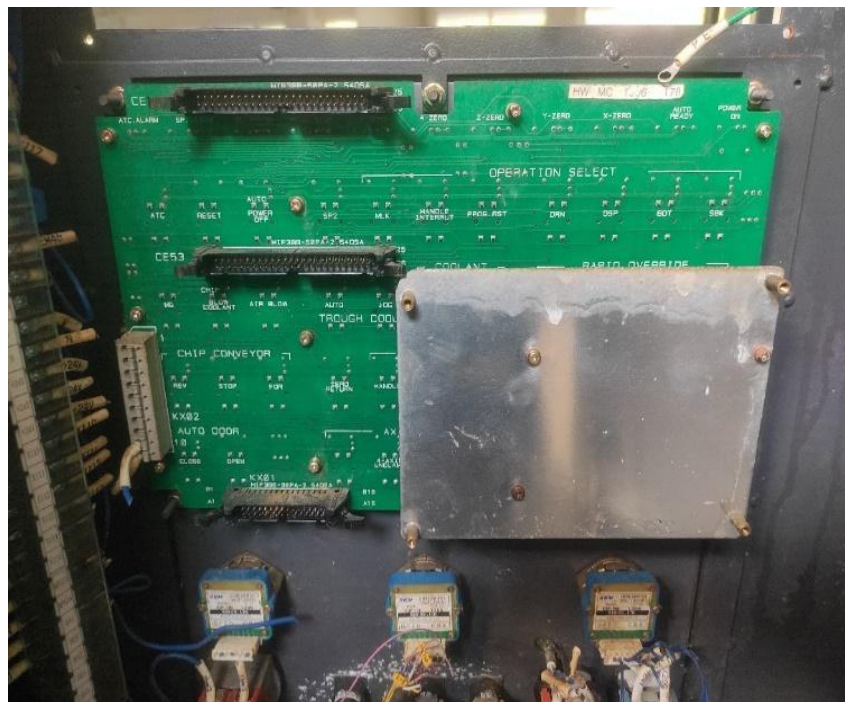


Figure 10 : HMI part II